

Alexander Arias

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EDUCATION

University of California - San Diego
Bachelor of Science in Computer Science

San Diego, CA
Dec 2023

EXPERIENCE

Qualcomm

Engineering Technician - Operations Department

San Diego, CA
June 2025 - May 2026

- Supported validation and testing workflows for next-generation SoC platforms used in flagship mobile devices.
- Executed high-volume automated and manual test procedures across tens of thousands of devices spanning Snapdragon, PMIC, RF, digital, and more validation environments.
- Operated within Python/PyCharm-based testing environments to run, monitor, and troubleshoot SoC diagnostic and validation workflows.
- Collaborated with engineers to execute test plans, reproduce issues, verify fixes, and maintain validation reliability.
- Performed hardware/software troubleshooting and documented validation metrics, test results, and debugging data using internal engineering tools.
- Assisted with Python-based debugging and test script support while consistently exceeding monthly performance metrics in a high-throughput semiconductor environment.

University of California: San Diego

Computer Science Outreach Volunteer - Eta Kappa Nu Engineering Honor Society

San Diego, CA
Jan. 2023 - Dec. 2023

- Delivered STEM outreach activities introducing fundamental computer science concepts to elementary and high school students.
- Supported interactive coding and algorithm demonstrations for student engagement in computing topics.
- Mentored high school students on academic pathways in computer science and engineering.

Computer Science Tutor - CSE Department

Sep. 2022 - Dec. 2022

- Tutored students in C and Assembly for Computer Organization and Systems Programming coursework.
- Provided one-on-one and group instruction to reinforce core concepts in low-level programming and computer architecture.

PROJECTS

Custom ISA and CPU | *System Verilog, Python, C*

- Led a 3-person team to design and implement a custom 9-bit instruction set architecture and CPU targeting Forward Error Correction workloads.
- Designed and implemented core CPU logic in SystemVerilog, including instruction decoding, control flow, and execution behavior.

Nachos Operating System | *Java, C, Linux Commands*

- Contributed to a 4-person team project implementing core operating system modules, including synchronization, CPU scheduling, and memory management.
- Implemented paging-based memory subsystem and assisted with debugging thread and concurrency behavior during system testing.

Graphics Shadow Mapping and 3D Rendering Projects | *C++, C, OpenGL*

- Applied advanced graphics techniques such as fragment shaders, Mandelbrot fractals, and 3D rotation algorithms to build immersive visual scenes.
- Collaborated with a team of 2 to develop a 3D graphics rendering application featuring dynamic shadow mapping using OpenGL.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, C#, JavaScript, Assembly, SystemVerilog, HTML, CSS

Frameworks and Libraries: .NET, Node.js, JUnit

Development Tools & Debugging: Git, GitHub, GDB, Valgrind, Linux command-line tools

Operating Systems: Linux, Windows

Productivity & Collaboration Tools: Microsoft Office 365, Google Workspace, AI-assisted tools

Testing & Validation: Software Testing, Automated Testing, Manual Testing, Hardware Validation, Test Execution, Issue Reproduction, Fix Verification, Test Documentation